

Lecture 3.1 : Conservation theorems

★ We won't be proving the conservation of the various quantities. We will simply derive some consequences of Newtonian dynamics.

I Conservation of linear momentum :

→ The total linear momentum $\vec{p}$ of a particle is conserved when the total force on it is zero.

$$\vec{F}_{\text{total}} = \dot{\vec{p}} = 0 \Rightarrow \vec{p} \text{ is a vector constant in time.}$$

★ Since $\dot{\vec{p}} = 0$: It can be applied for each component of linear momentum.

i.e. let $\vec{s}$ be $\vec{F} \cdot \vec{s} = 0$: $\vec{F} \perp \vec{s} \Rightarrow \dot{\vec{p}} \cdot \vec{s} = \vec{F} \cdot \vec{s} = 0$

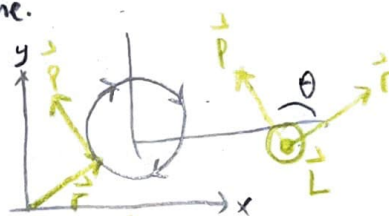
$$\Rightarrow \frac{d}{dt} [\vec{p} \cdot \vec{s}] = 0 \xrightarrow{\text{integrate}} \vec{p} \cdot \vec{s} = \text{Constant}$$

i.e. : component of linear momentum in a direction which the force vanishes is constant in time.

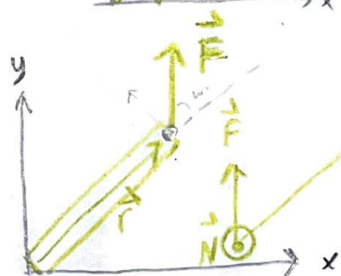
Angular momentum and torque :

$$\vec{L} \equiv \vec{r} \times \vec{p} : \text{angular momentum} \rightsquigarrow$$

$$\vec{N} \equiv \vec{r} \times \vec{F} : \text{torque} \rightsquigarrow$$



$$|\vec{L}| = |\vec{p}| |\vec{r}| \sin \theta$$



$$|\vec{N}| = |\vec{r}| |\vec{F}| \sin \theta$$

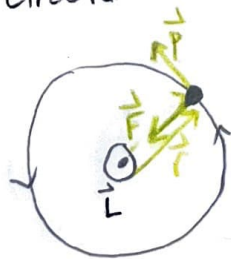
II Conservation of angular momentum

→ The angular momentum of a particle subject to no torque is conserved.

$$\vec{N} = \vec{r} \times \vec{F} = \vec{r} \times \dot{\vec{p}} ; \dot{\vec{L}} = \frac{d}{dt} (\vec{r} \times \vec{p}) = \dot{\vec{r}} \times \vec{p} + \vec{r} \times \dot{\vec{p}} ; \dot{\vec{r}} \times m\vec{v} = m\dot{\vec{r}} \times \dot{\vec{r}} = 0$$

→ $\dot{\vec{L}} = \vec{r} \times \dot{\vec{p}} = \vec{N}$: If no torque applied on a particle $\vec{L}$ is a vector constant in time.

e.g. Circular motion takes place in 2 dimensions. why?



$$\vec{N} = \vec{r} \times \vec{F} = 0 \Rightarrow \dot{\vec{L}} = 0 \Rightarrow \vec{L} = \text{constant}$$

Work and kinetic energy

as it moves

If a force $\vec{F}$ acts on a particle from point 1 to point 2, then the work done on this particle is defined as

$$W_{12} \equiv \int_1^2 \vec{F} \cdot d\vec{r} \rightarrow \text{WORK}$$

$d\vec{r}$: infinitesimal displacement vector
 $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

If $\vec{F}$ is the only force acting on the particle, we can write the integrand as

$$\vec{F} \cdot d\vec{r} = m \frac{d[\vec{v}]}{dt} \cdot \frac{d\vec{r}}{dt} dt = m \frac{d[\vec{v}]}{dt} \cdot \vec{v} dt = \frac{m}{2} \frac{d[\vec{v} \cdot \vec{v}]}{dt} dt = \frac{m}{2} \frac{d[v^2]}{dt} dt$$

$$= \frac{d}{dt} \left[\frac{1}{2} m v^2 \right] dt$$

$$\Rightarrow W_{12} = \int_1^2 \frac{d}{dt} \left[\frac{1}{2} m v^2 \right] dt = \int_{(1)}^{(2)} d \left[\frac{1}{2} m v^2 \right] = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2 = T_2 - T_1$$

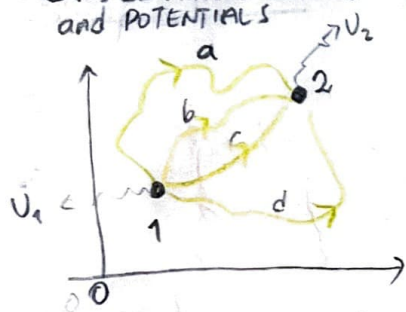
? Desc. is behind

$$T \equiv \frac{1}{2} m v^2 : \text{Kinetic energy} : [M] \frac{[L]^2}{[S]^2} \text{ S.I.: } \text{kg} \frac{\text{m}^2}{\text{s}^2} = \text{J} = \text{Joule}$$

★ If $T_1 > T_2 \Rightarrow W_{12} < 0$ particle has done work and its K.E. reduced.

3.2: Conservative forces and potentials

CONSERVATIVE FORCES: $\vec{F}$ has the property that the work required to move a particle from one position to another without any change in T depends only on the initial and final positions and not on the exact path taken by the particle. $\rightarrow \vec{F}$ is conservative!



★ Paths a, b, c, d are equivalent in terms of the work required to move a particle from 1 to 2.

So, if there are forces which satisfy the above property, we ^{simply} assume that $T_1 = T_2$. We still want $W_{12} \neq 0$. Therefore, there must be some other quantity which solely depend on particles coordinates. So that

$$W_{12} = \int_1^2 \vec{F} \cdot d\vec{r} \equiv U_1 - U_2 \quad \text{or in words} \quad \oint \vec{F} \cdot d\vec{r} = 0$$

★ We will call U the potential of the particle.

Let's assume that particle can only move in x direction.

$$W_{12} = \int_1^2 F_x dx = U_1 - U_2 : \text{This integral is satisfied only if } F_x = - \frac{dU}{dx}$$

Gradient operator.

$$\Rightarrow F_x = - \frac{dU}{dx} \quad \text{in 3D} \rightarrow \vec{F} = - \vec{\nabla} U \quad \vec{\nabla} \equiv \frac{d}{dx} \hat{i} + \frac{d}{dy} \hat{j} + \frac{d}{dz} \hat{k}$$

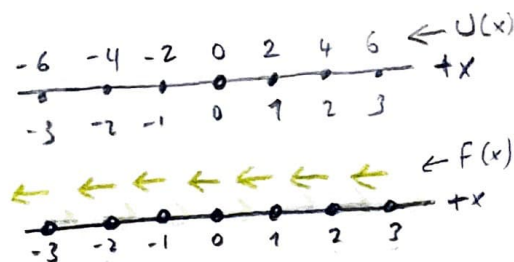
★ $\vec{\nabla}$ converts the scalar field U to a vectorial field $\vec{F}$.

★ U is defined only up to a constant: $\vec{\nabla} U \equiv \vec{\nabla} (U + c) \Rightarrow U$ has no absolute meaning on its own; only differences of potential are physically meaningful.

★ How $\vec{\nabla}$ converts scalar field to a vectorial field?

Let's assume that we are given $U(x) = 2x$

$$\Rightarrow \vec{F} = -\vec{\nabla} U = -\left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}\right) 2x = -2\hat{i}$$



It doesn't matter where the particle is there is always a constant force acting on it in $-\hat{i}$ direction

Example: Potential and potential energy

A particle of mass m is raised through a height h (by any path).

What is the work done on the particle?



$$W_{12} = \int_1^2 \vec{F} \cdot d\vec{r} = \int_0^h F_z dz = \int_0^h (mg) dz = +mg \int_0^h dz = +mgh$$

★ mgh is the work done by the force (or agent) which carries it to a height h .

★ If we asked what is the work done by the gravitation on the particle during its motion from 0 to h ?



$$W_{12} = \int_0^h \vec{F}_g \cdot d\vec{r} = - \int_0^h mg dz = -mgh$$

Like potentials and the potential energy, it is impossible to define an absolute K.E. for a particle. Both of these limitations are the result of the fact that selecting the origin of the coordinate system used to describe physical processes is always arbitrary.

LECTURE 3.2

Conservation of the total energy: (Give an example here)

! We assume that $\vec{F}$ is time independent

$$E \equiv T + U : \text{total energy}$$

$$\frac{dE}{dt} = \frac{dT}{dt} + \frac{dU}{dt} \xrightarrow[\text{consider}]{\text{to evaluate this}} \vec{F} \cdot d\vec{r} = d\left(\frac{1}{2} m v^2\right) = dT \Rightarrow \frac{dT}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt}$$

In most systems of interest, the potential energy is a function of position and time:

$$\Rightarrow U = U(\vec{r}, t) \Rightarrow \frac{dU}{dt} = \sum_i \frac{\partial U}{\partial x_i} \frac{dx_i}{dt} + \frac{\partial U}{\partial t} = \sum_i \frac{\partial U}{\partial x_i} \dot{x}_i + \frac{\partial U}{\partial t} = (\vec{\nabla} U) \cdot \frac{d\vec{r}}{dt} + \frac{\partial U}{\partial t}$$

$$\frac{dE}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt} + (\vec{\nabla} U) \cdot \frac{d\vec{r}}{dt} + \frac{\partial U}{\partial t} = \underbrace{(\vec{F} + \vec{\nabla} U)}_{-\vec{F}} \cdot \frac{d\vec{r}}{dt} + \frac{\partial U}{\partial t} = \frac{\partial U}{\partial t}$$

we assumed the total force is conservative.

If U is not an explicit function of time, then $dE/dt = 0$, i.e., the total energy is conserved.

Ⓣ The total energy of a particle in a conservative force field is a constant in time.

Energy as an equation of motion:

Consider a point particle under the influence of a conservative force with potential U . We know that the total energy $E = T + U$, which can also be called mechanical energy will not change in time.

$$E = T + U = \frac{1}{2} m v^2 + U(x) \rightarrow \text{we assumed for simplicity that } U \text{ is only a function of } x.$$

If we rearrange this equation:

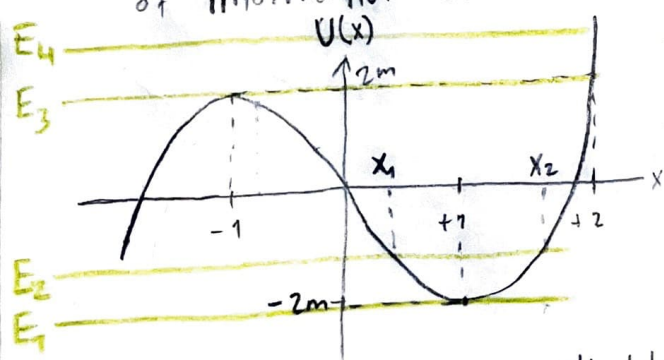
$$\frac{2}{m} [E - U] = v^2 \Rightarrow v = \pm \sqrt{\frac{2}{m} [E - U(x)]} = \frac{dx}{dt}$$
$$\Rightarrow \int_{t_0}^t dt = \int_{x_0}^x \frac{\pm dx}{\sqrt{\frac{2}{m} [E - U(x)]}} \Rightarrow t - t_0 = \int_{x_0}^x \frac{\pm dx}{\sqrt{\frac{2}{m} [E - U(x)]}} \rightarrow \text{we will turn back to this equation when we start studying oscillators}$$

Therefore, if we know $U(x)$, we can simply insert it into above equation and integrate it (by hand, or computer) to find $x(t)$! ★

What ^{does} the potential and total energy tell us? (U, x_0, v_0) defines every thing about a mechanical system.
 → This fixes the unit.
 → Emphasize that this is energy

Consider some potential $U(x) = m(x^3 - 3x)$

If we simply insert this into the eq. of motion obtained by the total energy we would have a crazy integral which has no solution in terms of the elementary functions. However even without solving the integral, $U(x)$ itself gives us a lot of information about the motion.



- First, notice the mathematical fact that $T \geq 0$ since $m \geq 0$ and $v^2 \geq 0$. Therefore, the only physically meaningful relationship between E and U is

$$T = E - U \geq 0 \Rightarrow E \geq U(x)$$

- This means that if the particle has a total energy E_1 the only physically possible position for its existence is $x = +1$. With E_1 , all the other points in the position space make no sense since they require the particle to have a negative kinetic energy, which is impossible. Similarly, for energies E_2 and E_3 the particle can only exist in the position intervals $[x_1, x_2]$ and $[-1, +2]$, respectively.

i) If we drop the particle at $+1$ with total energy E_1 , it will stay there forever since $F = -dU/dx = 0$ and $E = T + U \rightarrow -2m = T + 2m \Rightarrow T = 0$.

ii) If we drop the particle at $+1$ with total energy E_2 , $E = T + U \rightarrow E_2 = T - 2m$. Since $E_2 > -2m$, $T > 0$. It does not matter we don't know which direction ($+x$ or $-x$) the particle was going at the time we dropped it, we surely know that it will oscillate between $[x_1, x_2] \Rightarrow$ If it goes in $+x$ direction, it would be able to reach the position x_2 where it loses its all kinetic energy, which is completely converted to potential energy $U(x_2)$. At x_2 , the particle is exposed to a force $F = -\frac{dU(x)}{dx} \Big|_{x_2}$, which is in $-x$ direction. So the particle will similarly move until it reaches x_1 and the process repeats itself forever.

iii) If the particle has a total energy of E_2 and if we drop it at x_1 or x_2 , it would oscillate between $[x_1, x_2]$ again for the reasons indicated in ii).

iv) If particle has E_4 and if we simply drop it at point $+2$, it would go in $-x$ direction (first accelerating, then decelerating, and then accelerating again) forever.

v) If particle has E_4 and if we drop it some other point $x < +2$
 $\rightarrow$ If it initially goes in $+x$ direction, it would bounce from point $+2$ and go in $-x$ forever.
 $\rightarrow$ If it initially goes in $-x$ direction, it would simply keep on going in $-x$ forever.

vi) If particle has E_3 and we drop it at -1 , $T=0$ and $F = \frac{dU}{dx} \Big|_{x=-1} = 0$
 So, it will stay there forever.

vii) If particle has E_3 and we drop it at $+2$, the particle first falls into the dip and starts to climb up towards $x = -1$. It can never stop before it reaches $x = -1$ because before $x = -1$ it will still have some kinetic energy. But it also cannot arrive $x = -1$ with any excess kinetic energy. The only option is that the particle moves towards $x = -1$ at an ever decreasing speed, only reaching the maximum at time $t \rightarrow \infty$.
 To see this let's consider the Taylor expansion of $V(x)$ at $x \approx -1 + \epsilon$ with $\epsilon \ll 1$:

$$\begin{aligned} V(x) &= V(-1) + \left. \frac{dV(x)}{dx} \right|_{x=-1} (x+1) + \frac{1}{2!} \left. \frac{d^2V(x)}{dx^2} \right|_{x=-1} (x+1)^2 + \dots \\ &= m(-1+3) + m(3x^2-3) \Big|_{x=-1} (x+1) + \frac{1}{2} 6x \Big|_{x=-1} m(x+1)^2 + \dots \\ &= 2m + 0 + 3m(x+1)^2 + \dots = 2m - 3m(-1+\epsilon+1)^2 + \dots = 2m - 3m\epsilon^2 + \dots \end{aligned}$$

Since $\epsilon \ll 1$, let's drop ϵ^3 and higher terms:

$$V(x \approx -1 + \epsilon) \approx 2m - 3m\epsilon^2 \rightarrow V(\epsilon) = 2m - 3m\epsilon^2 : \text{function of } \epsilon$$

Now, we can use $x = -1 + \epsilon \Rightarrow dx = d\epsilon$

$$t - t_0 = \int_{\epsilon_0}^{\epsilon} \frac{1 d\epsilon}{\sqrt{\frac{2}{m} [2m - 3m\epsilon^2]}} = - \int_{\epsilon_0}^{\epsilon} \frac{d\epsilon}{\sqrt{6} \epsilon} = - \frac{1}{\sqrt{6}} \log\left(\frac{\epsilon}{\epsilon_0}\right)$$

When $\epsilon \rightarrow 0$, $x \rightarrow -1$, $t - t_0 \rightarrow +\infty$ *It takes infinite time to reach the top

Stable, unstable and neutral equilibrium points

$x = -1$: Unstable equilibrium
 $x = +1$: stable equilibrium } Ball inside/surface of the bowl example



Cases for $V(x) = m(x^2 - 3x)$ in summary:

① m is needed to fix units

- i) E_1 and $+1$
- ii) E_2 and $+1$
- iii) E_2 and x_1 or x_2
- iv) E_4 and $+2$
- v) E_4 and $x < +2$
- vi) E_3 and $x = -1$
- vii) E_3 and $x = +2$

if it goes $m+x$
if it goes $m-x$

$$1 \text{ J} = 1 \text{ kg} \frac{\text{m}^2}{\text{s}^2}$$

$$U(x) = m(x^2 - 3x)$$

$\downarrow$
 $\text{kg} \frac{\text{m}^2}{\text{s}^2} \quad N$

It can never stop before $x = -1$ since it will still have some T.

But also it cannot arrive $x = -1$ with any excess T.

★ Particle moves towards $x = -1$ with an ever decreasing speed, and only reaches there at time $t \rightarrow \infty$.

$$U(x) = U(-1) + \frac{dU(-1)}{dx}(x+1) + \frac{1}{2!} \frac{d^2U(-1)}{dx^2}(x+1)^2 + \dots$$

$$= m(1+3) + m(3-3)(x+1) + \frac{1}{2} m(-6)(x+1)^2 + \dots$$

$$= +2m + 0 - 3m(x+1)^2 + \dots$$

$$= 2m - 3m(x+1)^2 + \dots$$

$$\frac{dU}{dx} = 3x^2 - 3$$

$$\frac{d^2U}{dx^2} = 6x$$

$$U(-1) = 2m \quad \checkmark$$

Let us consider what is happening at $x = -1 + \epsilon$ where $\epsilon \ll 1$.

$$U(-1+\epsilon) = 2m - 3m(-1+\epsilon+1)^2 + \dots = 2m - 3m\epsilon^2 + \dots$$

Since $\epsilon \ll 1$, we can neglect the higher order terms:

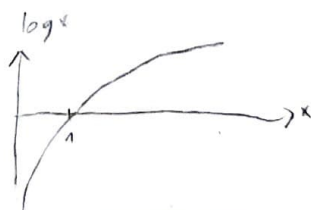
$$U(-1+\epsilon) \approx 2m - 3m\epsilon^2 \quad x = -1 + \epsilon \Rightarrow dx/d\epsilon = 1 \Rightarrow dx = d\epsilon$$

$$t_1 - t_0 = \int_{x_0}^{x_1} \frac{dx}{\sqrt{\frac{2}{m}(E - U)}} \rightarrow \int_{\epsilon_0}^{\epsilon_1} \frac{d\epsilon}{\sqrt{\frac{2}{m}(2m - 2m + 3m\epsilon^2)}} = \int_{\epsilon_0}^{\epsilon_1} \frac{d\epsilon}{\sqrt{6} \epsilon}$$

$$= \frac{1}{\sqrt{6}} \int_{\epsilon_0}^{\epsilon_1} \frac{d\epsilon}{\epsilon} = \frac{1}{\sqrt{6}} \log \epsilon \Big|_{\epsilon_0}^{\epsilon_1} = \frac{1}{\sqrt{6}} \log \left(\frac{\epsilon_1}{\epsilon_0} \right)$$

$$\Rightarrow t_1 - t_0 = \frac{1}{\sqrt{6}} \log \left(\frac{\epsilon_1}{\epsilon_0} \right)$$

$$\text{when } \epsilon_1 \rightarrow 0 \quad x \rightarrow -1 \quad t_1 - t_0 \rightarrow \infty$$



Stable: The ball can roll inside the bowl but it will eventually return to the bottom

Unstable: If we turn the bowl upside down and place the ball precisely outside in the middle it will stay there. However, if it slightly moves right or left it will not return to the middle forever.

Mathematical definition of stable and unstable equilibrium points

$U(x) = U_0 + x \left. \frac{dU}{dx} \right|_0 + \frac{x^2}{2!} \left. \frac{d^2U}{dx^2} \right|_0 + \frac{x^3}{3!} \left. \frac{d^3U}{dx^3} \right|_0 + \dots$

$\nearrow 0$: by assumption (we can always define a reference frame making this true).

We assumed that $x=0$ is the equilibrium point, we know that if $x=0$ is the equilibrium point:

$$\left. \frac{dU}{dx} \right|_0 = 0$$

$$\Rightarrow U(x) = \frac{x^2}{2!} \left. \frac{d^2U}{dx^2} \right|_0 + \frac{x^3}{3!} \left. \frac{d^3U}{dx^3} \right|_0 + \dots$$

Near the equilibrium $x \ll 0$
so we intentionally neglect all the terms except the first

$$U(x) \approx \frac{x^2}{2!} \left. \frac{d^2U}{dx^2} \right|_0$$

$$\Rightarrow \left. \frac{d^2U}{dx^2} \right|_0 > 0 \text{ Stable equilibrium}$$

$$\left. \frac{d^2U}{dx^2} \right|_0 < 0 \text{ unstable equilibrium}$$

Example: $V(x) = m(x^3 - 3x)$

$$\frac{dV}{dx} = m(3x^2 - 3)$$

$$\frac{d^2V}{dx^2} = 6mx$$

$$\left. \frac{dV}{dx} \right|_{-1} = 0$$

$$\left. \frac{d^2V}{dx^2} \right|_{-1} = -6m \rightarrow \text{Unstable equilibrium}$$

$$\left. \frac{dV}{dx} \right|_{+1} = 0$$

$$\left. \frac{d^2V}{dx^2} \right|_{+1} = 6m \rightarrow \text{Stable equilibrium}$$

-1 and +1 are equilibrium points

$$F = -\frac{dU}{dx} \rightarrow U(x) = -Fx + C \quad \text{we can choose } C=0 \text{ for simplicity without losing any generality}$$

$$E = T + U = \frac{1}{2}mv^2 - Fx \Rightarrow v^2 = \frac{2}{m}(E + Fx)$$

$$v = \pm \sqrt{\frac{2}{m}(E + Fx)} \Rightarrow \frac{dx}{dt} = \pm \sqrt{\frac{2}{m}(E + Fx)} \Rightarrow \frac{dx}{\pm \sqrt{\frac{2}{m}(E + Fx)}} = dt$$

$$\Rightarrow t - t_0 = \int_{x_0}^x \frac{\pm dx'}{\sqrt{\frac{2}{m}(E + Fx')}} \quad \frac{2}{m}(E + Fx') = u'$$

$$\frac{du'}{dx'} = \frac{2}{m}F \rightarrow dx' = \frac{m}{2F} du'$$

$$\Rightarrow t - t_0 = \int_{\frac{2}{m}(E + Fx_0)}^{\frac{2}{m}(E + Fx)} \frac{\pm \frac{m}{2F} du'}{u'^{1/2}} = \pm \frac{m}{2F} \int u'^{-1/2} du' = \pm \frac{m}{2F} \left[2u'^{1/2} \right]_{x_0}^x$$

$$= \pm \frac{m}{F} \left[\sqrt{\frac{2}{m}(E + Fx)} - \sqrt{\frac{2}{m}(E + Fx_0)} \right]$$

Choose $t_0 = 0 \rightarrow$ We started our the chronometer at $t_0 = 0$.

$$\pm \frac{F}{m} (t - t_0) + \sqrt{\frac{2}{m}(E + Fx_0)} = \sqrt{\frac{2}{m}(E + Fx)}$$

$$\pm t + \frac{\sqrt{2m}}{F} \sqrt{(E + Fx_0)} = \frac{\sqrt{2m}}{F} \sqrt{E + Fx}$$

↓ Square EVERYTHING

$$\Rightarrow \frac{2m}{F^2} (E + Fx) = t^2 \pm 2 \frac{\sqrt{2m}}{F} t \sqrt{(E + Fx_0)} + \frac{2m}{F^2} (E + Fx_0)$$

Plug in initial velocity: $E = \frac{1}{2}mv_0^2 + v(m) = \frac{1}{2}mv_0^2 - Fx_0 \Rightarrow E + Fx_0 = \frac{1}{2}mv_0^2$
 Total energy is directly found from initial conditions.

$$\frac{2m}{F^2} (E + Fx) = t^2 \pm 2 \frac{\sqrt{2m}}{F} t \sqrt{\frac{m}{2} v_0^2} + \frac{2m}{F^2} E + \frac{2m}{F^2} Fx_0$$

$$\cancel{\frac{2m}{F^2} E} + \cancel{\frac{2m}{F^2} Fx} = t^2 \pm 2 \frac{\sqrt{2m}}{F} t \sqrt{\frac{m}{2} v_0^2} + \cancel{\frac{2m}{F^2} E} + \cancel{\frac{2m}{F^2} Fx_0}$$

$$x(t) = \frac{F}{2m} t^2 \pm \frac{2F}{2m} \frac{\sqrt{2m}}{F} \sqrt{\frac{m}{2} v_0^2} t + \frac{2m}{F^2} Fx_0 \quad \text{initial}$$

$$x(t) = \frac{1}{2} \frac{F}{m} t^2 \pm v_0 t + x_0 \quad \pm \text{ includes the possible choices for direction.}$$

Ex: ROLLER COASTER PROBLEM (HARD)

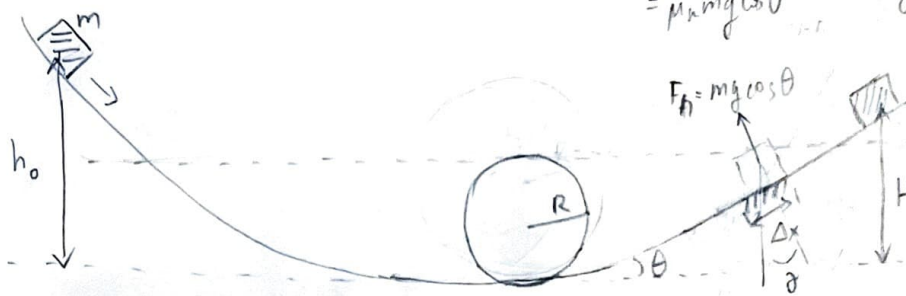
$$F_f = \mu_k N = \mu_k mg \cos \theta$$

$$mgh_o = mg2R + \frac{1}{2}mgR$$

$$= mgH + \mu_k mg \cos \theta \Delta x$$

We aren't asking what the block would reach. We want it to be there.

$$mgh_o = mgH + \mu_k mg \cos \theta \Delta x$$



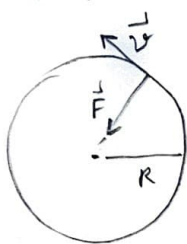
a) what is the minimum starting height is required such that the block just barely makes it through the top of the loop. (it doesn't lose contact at the top of the loop and reaches a height at least $H > 2R$ it)

b) Determine the speed of the block immediately after it emerges from the friction segment.

c) Now assume $H > R$. Is it possible that the block will not reach the top of the final hill, despite starting from that minimal h_o ?

Solution:

a) Eq. of motion for circ. motion



$$\frac{mv^2}{R} = ma$$

$$\frac{mv^2}{R} \geq mg \Rightarrow v^2 \geq gR$$

$$m \left(\frac{F_{\text{net}}}{\text{loop}} \right) = mg(2R) + \frac{1}{2}mgR$$

mgh_o : initial potential energy

$mg2R$: final potential energy

$mgh_o - mg2R$: change in potential energy

$$mgh_o - mg2R = \frac{1}{2}mv_{\text{top}}^2$$

$$gh_o - g2R = \frac{1}{2}v_{\text{top}}^2 \Rightarrow gh_o = \frac{1}{2}v_{\text{top}}^2 + g2R$$

$$\Rightarrow h_o = \frac{1}{2g}v_{\text{top}}^2 + 2R$$

$$h_{o,\text{min}} = \frac{1}{2g}gR + 2R = \frac{5R}{2}$$

Some of its KE must be broken away by the work done by friction

$$F_f =$$

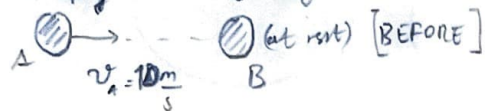
$$mgh_o - mgh_o$$

$$mg2R$$

Ex: INELASTIC COLLISION PROBLEM

$$m_A = 4 \text{ kg}$$

$$m_B = 5 \text{ kg}$$



$$4 \text{ kg} = 4 \text{ kg}$$

$$4 \times 10 \text{ kg} = 40 \text{ kg}$$

They collide inelastically and stick together.

a) Determine the velocity v after collision.

b) Compute K.E before and after collision.

How much K.E is lost?

d) If instead the collision was partially inelastic: they do not stick together.

but half of the K.E of the system is lost. Find the velocities A, B after the collision.

c) Suppose now that the combined block from part (a) hits a spring of spring constant $k = 1200 \frac{\text{N}}{\text{m}}$ and compresses it (without friction). How far the spring compress?

$$d) E_i = T_i = \frac{1}{2} m_A v_{Ai}^2$$

$$E_f = \frac{E_i}{2} \Rightarrow T_f = \frac{T_i}{2}$$

$$E_f = T_f = \frac{1}{2} m_A v_{Af}^2 + \frac{1}{2} m_B v_{Bf}^2$$

$$\Rightarrow \frac{1}{2} m_A v_{Af}^2 + \frac{1}{2} m_B v_{Bf}^2 = \frac{1}{4} m_A v_{Ai}^2$$

$$P_i = P_f$$

$$m_A v_{Ai} = m_A v_{Af} + m_B v_{Bf}$$

$$m_A v_{Af}^2 + m_B v_{Bf}^2 = \frac{1}{2} m_A v_{Ai}^2$$

2 eq. 2 unknowns

$$\frac{m_A v_{Ai}}{m_B} - \frac{m_A v_{Af}}{m_B} = v_{Bf}$$

$$m_A v_{Af}^2 + m_B \left(\frac{m_A v_{Ai}}{m_B} - \frac{m_A v_{Af}}{m_B} \right)^2 = \frac{1}{2} m_A v_{Ai}^2$$

$$\frac{m_A}{m_B} v_{Af}^2 + \frac{m_A}{m_B} (v_{Ai}^2 - 2 v_{Ai} v_{Af} + v_{Af}^2) = \frac{1}{2} m_A v_{Ai}^2$$

$$v_{Af}^2 + \frac{m_A}{m_B} v_{Ai}^2 - \frac{m_A}{m_B} 2 v_{Ai} v_{Af} + \frac{m_A}{m_B} v_{Af}^2 = \frac{v_{Ai}^2}{2}$$

$$\left(1 + \frac{m_A}{m_B} \right) v_{Af}^2 - \left(2 \frac{m_A}{m_B} v_{Ai} \right) v_{Af} + \left(\frac{m_A}{m_B} v_{Ai}^2 - \frac{v_{Ai}^2}{2} \right) = 0$$

$$\left(1 + \frac{m_A}{m_B} \right) v_{Af}^2 - \left(2 \frac{m_A}{m_B} v_{Ai} \right) v_{Af} + \left(\frac{m_A}{m_B} - \frac{1}{2} \right) v_{Ai}^2 = 0$$

$$v_{Af} = - \frac{2 m_A}{m_B} v_{Ai} + \sqrt{\left(\frac{2 m_A}{m_B} \right)^2 - 4 \left(1 + \frac{m_A}{m_B} \right)}$$

Impulse: A vector quantity showing the change of momentum of an object.

$$\vec{J} \equiv \vec{F} \Delta t \quad \left(\text{in SI } \text{kg} \frac{\text{m}}{\text{s}} \right)$$

★ It is actually a different interpretation of Newton's 2nd law in finite limit. (This approx. is exact for only constant forces. Otherwise it's just a linear approximation.)

$$\vec{F} \approx \frac{\Delta \vec{p}}{\Delta t} \Rightarrow \vec{F} \Delta t \approx \Delta \vec{p} = \vec{p}_2 - \vec{p}_1$$

Power: A quantity measuring the rate at which work is done or the rate the energy is transformed/transferred. (How much energy is delivered per unit time?)

$$P \equiv W/t \quad \text{in SI } \frac{1 \text{ J}}{\text{s}} \equiv 1 \text{ W.}$$

or $P \equiv E/t$

- Power quantifies how quickly work is done or how quickly the energy is transformed.

(i) Light bulb: 5W $\rightarrow$ It turns 5J of electric energy into "light energy" and heat in 1s.

(ii) 1 horse power: (unit of power) (it is actually defined in imperial units)

The power to ~~move~~ ^{lift} a mass of 75 kilograms against the Earth's gravitational force over a distance of 1m. in one second.

Let's convert it to watts:

$$\sim mgh = 75 \times 10 \times 1 \text{ kg} \frac{\text{m}}{\text{s}^2} \text{m} = 750 \text{ J}$$

$$\Rightarrow 1 \text{ HP} \equiv 750 \text{ W} = 0.75 \text{ kW} \rightarrow \text{This is } 98.6\% \text{ of an imperial horsepower}$$

(DIN)

According to a German ^{non-profit} ~~standardization~~ ^{national} organization

★ James Watt came up with this idea to compare the output of his steam engines to work rate of horses.

Lifting 550 pounds one foot in one second

Examples for impulse



Problem 1: A soccer player kicks a ball with a force of 200 N. The ball has a mass of 0.5 kg, and the force is applied for 0.1 seconds. Calculate the change in momentum and the final velocity of the ball.

Solution 1: First, calculate the impulse using the formula $J = F \cdot \Delta t$, where F is the force and Δt is the time.

$$J = 200 \text{ N} \cdot 0.1 \text{ s} = 20 \text{ N}\cdot\text{s}$$

Now, use the impulse-momentum theorem $\Delta p = J$ to find the change in momentum:

$$\Delta p = 20 \text{ N}\cdot\text{s}$$

Next, calculate the final velocity using the formula $p = mv$, where p is momentum and m is mass:

$$\Delta p = mv_f - mv_i$$

$$20 \text{ N}\cdot\text{s} = (0.5 \text{ kg}) \cdot v_f - (0.5 \text{ kg}) \cdot 0 \text{ m/s (initial velocity)}$$

Now, solve for v_f :

$$v_f = (20 \text{ N}\cdot\text{s} + 0.5 \text{ kg} \cdot 0 \text{ m/s}) / 0.5 \text{ kg} = 40 \text{ m/s}$$

So, the change in momentum is 20 N·s, and the final velocity of the ball is 40 m/s.



Problem 2: A ^{block of mass} car crashes into a wall and comes to a stop in 0.2 seconds. If the car has a mass of 1,200 kg, what is the average force exerted on the car during the collision?

Solution 2: First, calculate the change in momentum using the impulse-momentum theorem $\Delta p = J$. We know the car comes to a stop, so the initial momentum (mv_i) is equal to the final momentum (0, as the velocity becomes 0).

$$\Delta p = 0 - (1,200 \text{ kg}) \cdot v_i$$

Now, we need to calculate the impulse:

$$\Delta p = J = F \cdot \Delta t$$

$$0 - (1,200 \text{ kg}) \cdot v_i = F \cdot 0.2 \text{ s}$$

Now, solve for F:

$$F = (1,200 \text{ kg} \cdot v_i) / 0.2 \text{ s}$$

Since the car's initial velocity v_i is unknown, you would need that value to calculate the force.

Problem 3: A rocket with a mass of 1,000 kg is launched vertically. It expels exhaust gases with a speed of 1,000 m/s, generating an upward force of 200,000 N for 10 seconds. Calculate the rocket's final velocity.

Solution 3: First, calculate the impulse using the formula $J = F \cdot \Delta t$, where F is the force and Δt is the time:

$$J = (200,000 \text{ N}) \cdot (10 \text{ s}) = 2,000,000 \text{ N}\cdot\text{s}$$

Now, use the impulse-momentum theorem $\Delta p = J$ to find the change in momentum:

$$\Delta p = 2,000,000 \text{ N}\cdot\text{s}$$

Next, calculate the initial momentum (mv_i) using the rocket's mass and the exhaust velocity:

$$mv_i = (1,000 \text{ kg}) \cdot (-1,000 \text{ m/s}) = -1,000,000 \text{ kg}\cdot\text{m/s}$$

Now, solve for the final momentum (mv_f):

$$\Delta p = mv_f - mv_i$$

$$2,000,000 \text{ N}\cdot\text{s} = (1,000 \text{ kg}) \cdot v_f - (-1,000,000 \text{ kg}\cdot\text{m/s})$$

Now, solve for v_f :

$$v_f = (2,000,000 \text{ N}\cdot\text{s} + 1,000,000 \text{ kg}\cdot\text{m/s}) / (1,000 \text{ kg}) = 3,000 \text{ m/s}$$

So, the rocket's final velocity is 3,000 m/s.

Examples for power

Problem 1: A machine does 2,000 joules of work in 5 seconds. Calculate the power output of the machine.

Solution 1: To calculate power, use the formula $P = W / t$, where W is the work done and t is the time.

$$P = 2,000 \text{ joules} / 5 \text{ seconds} = 400 \text{ watts (W)}$$

So, the power output of the machine is 400 watts.

Problem 2: A car engine generates 300 horsepower. Convert this power into kilowatts.

Solution 2: 1 horsepower (hp) is equivalent to 0.7457 kilowatts (kW).

$$P \text{ (in kW)} = 300 \text{ hp} * 0.7457 \text{ kW/hp} = 223.71 \text{ kW}$$

So, the power output of the car engine is approximately 223.71 kilowatts.

Problem 3: A person climbs a flight of stairs that is 20 meters high in 10 seconds. Calculate the average power output of the person during the climb. Assume the person has a mass of 70 kg.

Solution 3: First, calculate the work done by the person using the formula $W = mgh$, where m is mass, g is the acceleration due to gravity (approximately 9.81 m/s^2), and h is the height.

$$W = (70 \text{ kg}) * (9.81 \text{ m/s}^2) * (20 \text{ m}) = 13,674 \text{ joules}$$

Now, calculate the power using $P = W / t$, where t is the time taken.

$$P = 13,674 \text{ joules} / 10 \text{ seconds} = 1,367.4 \text{ watts (W)}$$

So, the average power output of the person during the climb is 1,367.4 watts.

Problem 4: A wind turbine generates 500,000 joules of electrical energy in 10 seconds. Calculate the power output of the wind turbine.

Solution 4: To calculate power, use the formula $P = W / t$, where W is the energy generated and t is the time.

$$P = 500,000 \text{ joules} / 10 \text{ seconds} = 50,000 \text{ watts (W)}$$

So, the power output of the wind turbine is 50,000 watts, or 50 kilowatts (kW).

Problem 5: A microwave oven consumes 1,200 watts of power while heating food for 3 minutes. Calculate the total energy consumed during this time.

Solution 5: First, convert the time to seconds:

$$3 \text{ minutes} = 3 * 60 \text{ seconds} = 180 \text{ seconds}$$

Now, calculate the energy consumed using the formula $E = P * t$, where P is power and t is time.

$$E = (1,200 \text{ watts}) * (180 \text{ seconds}) = 216,000 \text{ joules}$$

So, the microwave oven consumes 216,000 joules of energy while heating the food.

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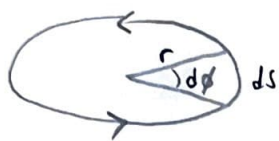
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So, the microwave oven consumes 216,000 joules of energy while heating the food.

Angular velocity : Angular velocity is a useful definition to describe rotating particles and bodies.



$$\omega \equiv \frac{d\phi}{dt} [T^{-1}] : SI : 1/s \quad \phi \text{ is defined in radians}$$

1 radian $\rightarrow$



Length of the circular arc is equal to the radius of the circle.

angular velocity in terms of linear velocity and radius.

Relationship btw. linear vel. and angular velocity $\rightarrow$

$$ds = r d\phi \quad \left(\frac{ds}{dt} \right) = r \left(\frac{d\phi}{dt} \right)$$

linear velocity $\qquad$ angular velocity

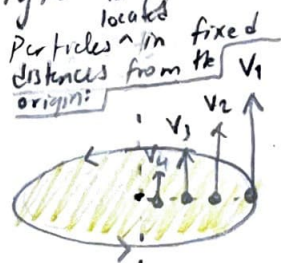
$$v = r\omega \rightarrow \omega = v/r$$

Rotational inertia of a system of particles

Kinetic energy is associated with the linear velocity of the bodies:

$$T = \frac{1}{2} m v^2$$

If we consider a rigid body rotating around an axis, this definition of T is not very useful because different points with different distances to the rotation axis on the rigid body will have different linear velocities. Therefore, we cannot assign a unique kinetic energy for the given rigid body. What we need is to define the kinetic energy of the body with a fixed parameter valid for all points on the rigid body.



$$v_1 > v_2 > v_3 > v_4$$

However, the disc has a unique angular velocity which is invariant for all parts of the body.

Therefore:

$$T = \sum_{i=1}^N \frac{1}{2} m_i v_i^2 = \sum_{i=1}^N \frac{1}{2} m_i (\omega_i^2 r_i^2) \rightarrow \text{we know that } \omega_i = \omega$$

$$= \frac{1}{2} \omega^2 \left(\sum_{i=1}^N m_i r_i^2 \right) = \frac{1}{2} I \omega^2$$

I $\rightarrow$ rotational inertia.

I : It is a property combining the mass and distribution of the particles around the rotation axis for a rigid body.

Relationship between L, W, I → angular momentum
→ rotational inertia
→ angular velocity

$$L = r p = r m v = r m r \omega = \underbrace{r^2 m}_{I} \omega : \text{angular momentum of a single particle}$$

$$L_{\text{TOTAL}} = \omega \sum_{i=1}^N m_i r_i^2 = \omega I$$

→ for a "rigid" rotating system of particles

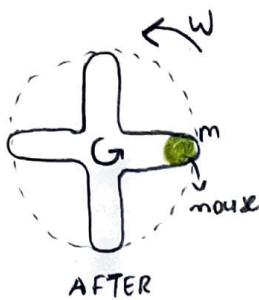
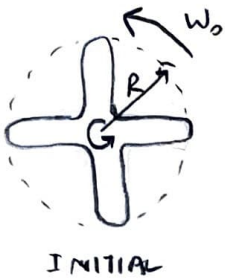
$$L_{\text{TOTAL}} = I \omega$$

$$L = I \omega$$

↓ analogy

$$p = m v$$

Example 2) Conservation of angular momentum



A mouse of mass m jumps on the outside edge of a freely turning ceiling fan of rotational inertia I and radius R . By what ratio does the angular velocity change?

$$L_{\text{initial}} = L_{\text{final}}$$

$$I \omega_0 = I \omega + R m v$$

$$= I \frac{v}{R} + R m v = \frac{v}{R} (I + m R^2)$$

$$\Rightarrow I \frac{v_0}{R} = \frac{v}{R} (I + m R^2) \Rightarrow \frac{v}{v_0} = \frac{I}{I + m R^2} \Rightarrow \frac{\omega}{\omega_0} = \frac{I}{I + m R^2}$$

Example 1) An ice skater spins with their arms out each side. In this position, they have a moment of inertia 50 kgm^2 , and rotate with an angular velocity of 4.5 rad/s . Then, the ice skater bring their arms closer to their chest to spin faster, decreasing their mom. of inertia to 40 kgm^2 , what is their new angular velocity?

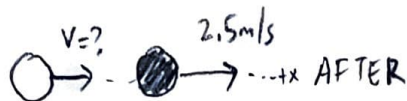
$$L_{\text{initial}} = L_{\text{final}}$$

$$I_i \omega_i = I_f \omega_f \rightarrow \omega_f = \frac{I_i}{I_f} \omega_i = \frac{50}{40} \cdot 4.5 \frac{\text{rad}}{\text{s}} = 5.6 \frac{\text{rad}}{\text{s}} \quad (\text{not significant})$$

Exemple 1) Cons. of lin. mom.

of 15g

Two hockey pucks of identical mass are on a flat, horizontal ice hockey rink. Black puck is motionless; the white is moving 2.5 m/s. It collides with the motionless black puck. After collision black puck moves with 2.5 m/s. What is the velocity of white after the collision?



$$\vec{P}_{\text{final}} = \vec{P}_{\text{initial}}$$

The motion takes place in 1D (say x direction)

$$\rightarrow P_{x,i} = P_{x,f}$$

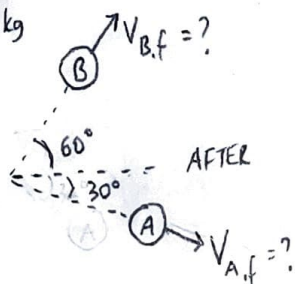
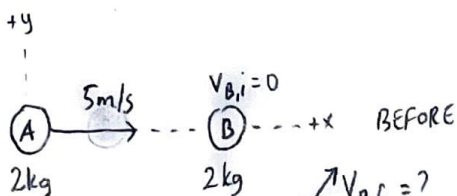
$$m_w V_{w,i} + m_b V_{b,i} = m_w V_{w,f} + m_b V_{b,f}$$

$$\begin{cases} m_w = m_b \\ V_b = 0 \end{cases}$$

$$\Rightarrow V_{w,i} = V_{w,f} + V_{b,f} \quad 2.5$$

$$= V_{w,f} = 0$$

Example 2) Cons. of lin. mom.

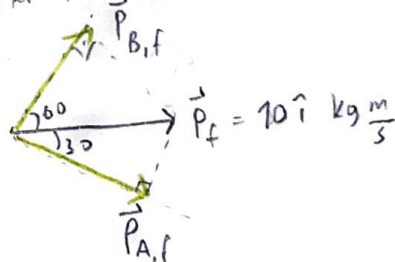


$$\vec{P}_i = \vec{P}_f$$

Before collision the total momentum in +x direction and it is

$$5.2 \text{ kg m/s} = 10 \text{ kg m/s}$$

Therefore, after the collision the total momentum has to be again 10 kg m/s in +x direction. So, all we need to do is to find the components of the momentum vector in particle A and B direction after the collision.



$$|P_{A,f}| = 10 \cos 30 = 10 \frac{\sqrt{3}}{2} = 5\sqrt{3} \text{ kg m/s}$$

$$|P_{B,f}| = 10 \cos 60 = 10 \frac{1}{2} = 5 \text{ kg m/s}$$

$$\Rightarrow |V_{A,f}| = \frac{5\sqrt{3}}{2} \frac{\text{m}}{\text{s}} \quad |V_{B,f}| = \frac{5}{2} \frac{\text{m}}{\text{s}}$$

Example 3:

10.10.22

$$b) \quad \frac{1}{2} m v^2 - \mu_k m g d \rightarrow \frac{1}{2} k x^2 + \mu_k m g x$$

$$\frac{1}{2} k x^2 + \mu_k m g x = \frac{1}{2} m v^2 - \mu_k m g d$$

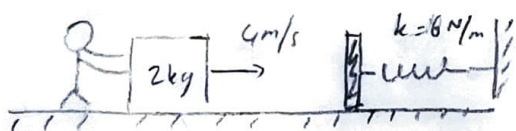
$$\underbrace{\left(\frac{1}{2} k\right)}_A x^2 + \underbrace{(\mu_k m g)}_B x - \underbrace{\left(\frac{1}{2} m v^2 + \mu_k m g d\right)}_C = 0$$

$$A x^2 + B x + C = 0$$

$$x_{1,2} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

$$= \frac{-\mu_k m g \pm \sqrt{\mu_k^2 m^2 g^2 - 2 k \left(-\frac{1}{2} m v^2 + \mu_k m g d\right)}}{2 \cdot \frac{1}{2} k}$$

$$= -\frac{\mu_k m g}{k} \pm \sqrt{\frac{\mu_k^2 m^2 g^2}{k^2} + \frac{m v^2}{k} - \frac{\mu_k m g d}{k}} \approx 1.1 \text{ m}$$



a) Slippery surface?

b) Rough floor $\mu_k = 0.2$?

$$a) \quad \frac{1}{2} m v^2 = \frac{1}{2} k x^2 \rightarrow x = \sqrt{\frac{m}{k} v^2} = \sqrt{\frac{m}{k}} v \approx 2.3 \text{ m}$$

Example 4: A particle of mass m has speed $v = \alpha/x$, where x is displacement. Find the force $F(x)$ responsible.

$$F = m a = m \frac{dv}{dt} = m \frac{dv}{dx} \frac{dx}{dt} = m v \left(\frac{dv}{dx} \right)$$

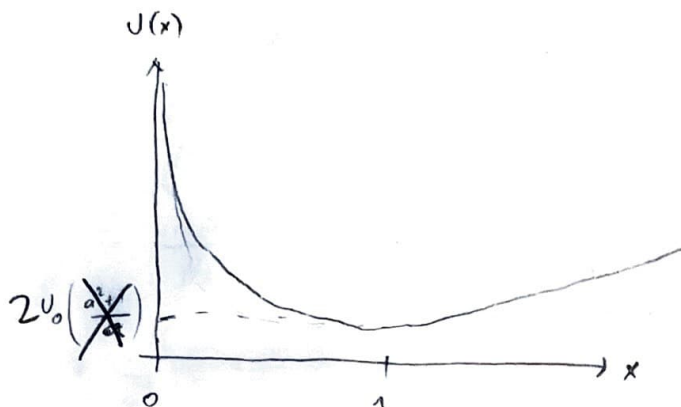
$$= m \alpha \frac{1}{x} (-\alpha) \frac{1}{x^2} = -m \frac{\alpha^2}{x^3}$$



Example 5: Consider a particle moving in the region $x > 0$ under the influence of the potential

$$U(x) = U_0 \left(\frac{a}{x} + \frac{x}{a} \right)$$

where $U_0 = 1\text{J}$ and $a = 2\text{m}$. Plot the potential, find the equilibrium points, and determine whether they are maxima or minima.



$$\frac{dU}{dx} = U_0 \left(-\frac{a}{x^2} + \frac{1}{a} \right)$$

$$U_0 \left(-\frac{a}{x^2} + \frac{1}{a} \right) = 0$$

$$\frac{1}{a} = \frac{a}{x^2}$$

$$x^2 = a^2$$

$$x = \pm a \rightarrow \text{we only consider } x = a$$

since $x < 0$ has no meaning.

$$\frac{d^2U}{dx^2} = 2ax^{-3}$$

$$\left. \frac{d^2U}{dx^2} \right|_{x=a} = \frac{2a}{a^3} = \frac{2}{a^2} > 0$$

Concave up

minima
stable eq.

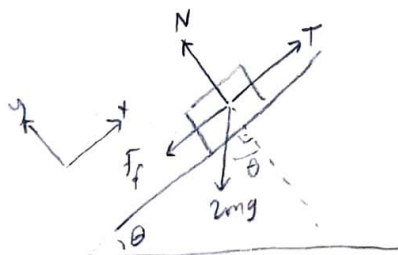
Example 6:

{Let's assume that system is moving in this direction}

Two blocks of unequal mass are connected by a string over a smooth pulley. If the coefficient of kinetic friction is μ_k what angle θ_0 of the incline allows the masses to move at a constant speed?

Forces on the hanging mass

The forces on the other mass:



$$\begin{aligned} \text{y-dir.} \\ N - 2mg \cos \theta &= m\ddot{y} = 0 \\ N &= 2mg \cos \theta \end{aligned}$$



$$mg - T = ma$$

$$\Rightarrow T = m(g - a)$$

$$\text{x-dir.} \\ F_f = \mu_k N = 2\mu_k mg \cos \theta$$

$$T - 2mg \sin \theta - 2\mu_k mg \cos \theta = ma$$

$$m(g - a) - 2mg \sin \theta - 2\mu_k mg \cos \theta = ma$$

$$g - 2g \sin \theta - 2\mu_k g \cos \theta = 2a$$

when $\theta = \theta_0$ (allowing constant speed) $a = 0$

$$g - 2g \sin \theta_0 - 2\mu_k g \cos \theta_0 = 0$$

$$g = 2g \sin \theta_0 + 2\mu_k g \cos \theta_0$$

$$\frac{1}{2} = \sin \theta_0 + \mu_k \cos \theta_0$$

$$\frac{1}{2} = \sin \theta_0 + \mu_k (1 - \sin^2 \theta_0)^{1/2}$$

$$\frac{1}{2} = \sin \theta_0 + \mu_k \sqrt{1 - \sin^2 \theta_0}$$

$$\frac{1}{\mu_k} \left(\frac{1}{2} - \sin \theta_0 \right) = \sqrt{1 - \sin^2 \theta_0}$$

$$\frac{1}{\mu_k^2} \left(\frac{1}{2} - \sin \theta_0 \right)^2 = 1 - \sin^2 \theta_0$$

$$\frac{1}{\mu_k^2} \left(\frac{1}{4} - 2 \frac{1}{2} \sin \theta_0 + \sin^2 \theta_0 \right) = 1 - \sin^2 \theta_0$$

$$\frac{1}{4\mu_k^2} - \frac{\sin \theta_0}{\mu_k^2} + \frac{\sin^2 \theta_0}{\mu_k^2} = 1 - \sin^2 \theta_0$$

$$\frac{1}{4\mu_k^2} - \frac{\sin \theta_0}{\mu_k^2} + \frac{\sin^2 \theta_0}{\mu_k^2} + \sin^2 \theta_0 = 1$$

$$\left(1 + \frac{1}{\mu_k^2} \right) \sin^2 \theta_0 - \frac{1}{\mu_k^2} \sin \theta_0 + \frac{1}{4\mu_k^2} - 1 = 0$$

$$ax^2 + bx + c = 0$$

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sin \theta_0 = \frac{+\frac{1}{\mu_k^2} \pm \sqrt{\frac{1}{\mu_k^4} - 4 \left(1 + \frac{1}{\mu_k^2} \right) \left(\frac{1}{4\mu_k^2} - 1 \right)}}{2 \left(1 + \frac{1}{\mu_k^2} \right)}$$

Exercise 7)

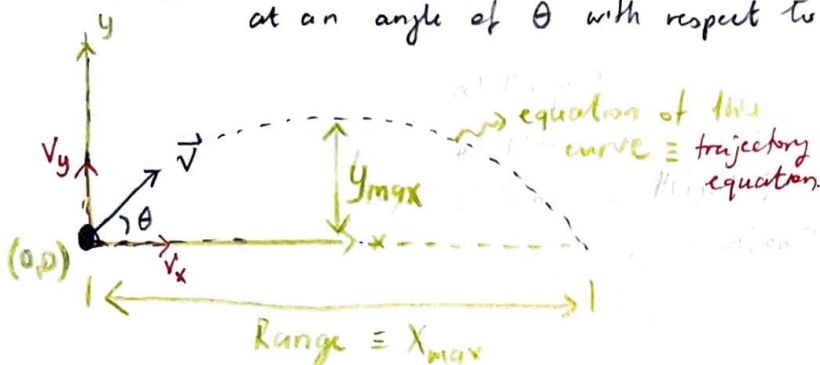
A train moves at a constant speed u . A woman on the train throws a ball of mass m straight ahead with a speed v with respect to herself. What is the K.E. gain of the ball as measured by the person on the train? (b) by a person standing by the railroad track? (c) How much work is done by the woman throwing the ball and (d) by the train?

Exercise 8) Moment of inertia of a disc on cylinder.

Projectile Motion.

Describes the motion of an object launched into the air. It occurs when an object is given an initial velocity and is subject only to the force of gravity (air resistance is neglected)

Example: A point object is launched in the air at an angle of θ with respect to the horizontal.



- Time of flight (t_{flight})
- Maximum height (y_{max})
- Range (x_{max})
- Trajectory equation

★ It is crucial to consider the motion in terms of velocity's x and y components

In y-direction:

$$-v_y = \frac{dv}{dt}$$

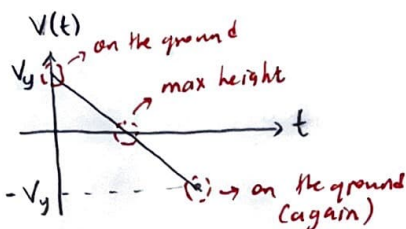
$$-g = \frac{dv}{dt}$$

$$-gt + v_y = v(t)$$

$$-\frac{1}{2}gt^2 + v_y t + y_0 = y(t)$$

$\uparrow$ 0 (by assumption)

$$y\left(\frac{t_{\text{flight}}}{2}\right) = -\frac{1}{2}g\frac{v_y^2}{g^2} + v_y \frac{v_y}{g} = \frac{v_y^2}{g} - \frac{1}{2}\frac{v_y^2}{g} = \frac{1}{2}\frac{v_y^2}{g} = y_{\text{max}}$$



• when $v(t)$ is 0, the particle reaches its max height

$$\Rightarrow -gt + v_y = 0 \Rightarrow v_y = gt \Rightarrow \frac{t_{\text{flight}}}{2} = \frac{v_y}{g}$$

$$\Rightarrow t_{\text{flight}} = \frac{2v_y}{g}$$

$$y_{\text{max}} = \frac{1}{2} \frac{v_y^2}{g}$$

In x-direction:

$$x_{\text{max}} = v_x t_{\text{flight}} = \frac{2v_x v_y}{g}$$

$$x_{\text{max}} = \frac{2v_x v_y}{g}$$

$$\begin{aligned} v_x &= v \cos \theta \\ v_y &= v \sin \theta \end{aligned}$$

Trajectory equation: How y changes with respect to x?

$$y(t) = v_y t - \frac{1}{2}gt^2$$

$$x(t) = v_x t \rightarrow t = \frac{x}{v_x}$$

$$y = \frac{v_y}{v_x} x - \frac{1}{2}g \frac{x^2}{v_x^2}$$

a quadratic equation (a parabola)

• By using $v_x = v \cos \theta$ and $v_y = v \sin \theta$ we can simplify some of these expressions:

$$x_{\text{max}} = \frac{2v^2 \sin \theta \cos \theta}{g} = \frac{v^2 \sin(2\theta)}{g}$$

$$y = \frac{v \sin \theta}{v \cos \theta} x - \frac{1}{2}g \frac{x^2}{v^2 \cos^2 \theta} = \tan \theta x - \frac{g \sec^2 \theta}{2v^2} x^2$$

A numerical ex. for projectile motion:

A soccer player kicks a ball with an initial speed $\frac{20\text{ m/s}}{\sqrt{2}}$ at an angle 45° with respect to horizontal. Find the following:

- a) t_{flight} (flight time) b) y_{max} (max. height) c) x_{max} (range) (Assume $g = 10\text{ m/s}^2$)

$$a) t_{\text{flight}} = \frac{2}{g} V_y = \frac{2}{g} V \sin \theta = \frac{2}{10} \frac{20}{\sqrt{2}} \frac{\sqrt{2}}{2} \frac{\text{m}}{\text{s}} = 2\text{ s}$$

$$b) y_{\text{max}} = \frac{1}{2} \frac{V_y^2}{g} = \frac{1}{2g} V^2 \sin^2 \theta = \frac{1}{2 \times 10} \frac{400}{2} = 5\text{ m}$$

$$c) x_{\text{max}} = \frac{V^2 \sin(2\theta)}{g} = \frac{400}{10} \sin[2 \times 45] = 40\text{ m}$$

Friction

It is a force resists the relative motion or tendency of such motion of two surfaces in contact. It acts in the opposite direction of the movement or intended movement between two surfaces.

- Static friction: prevents two surfaces from moving relative to each other. It must be overcome to initiate motion. it exists when an object is stationary, its value increases up to a maximum before the object starts moving.

$$F_s \leq N$$

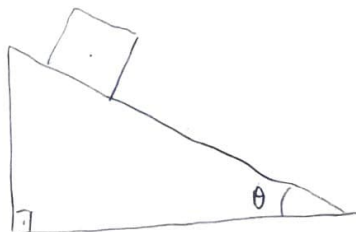
$$F_s = \mu_s N$$

(Dynamic)

- Kinetic friction: it is the frictional force acting between two surfaces in relative motion. Once the object starts moving kinetic friction takes over and its value is generally lower than that of static friction.

$$F_k = \mu_k N$$

Ex. Inclined plane with friction (μ_k)



Ex 6